



COMPANY STRATEGY

Building Durable Competitive Advantage

The 7 Powers Framework Applied to AI-Native Systems
Engineering

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The decision graph is the moat.

Luvian wins by building the **decision graph for safety-critical engineering**: a typed, queryable, auditable graph of engineering decisions and their consequences that no competitor covers at full depth.

The graph accumulates organizational intelligence that makes AI better with every interaction, creates switching costs that deepen with every trace link, and produces the compliance evidence that regulated buyers cannot do without.

4

Domains in one graph

Architecture + Requirements + Safety + V&V

0

Competitors cover all four

Structural differentiation

7

Powers to build

Helmer's framework for durable advantage

HAMILTON HELMER'S FRAMEWORK

The only 7 sources of durable advantage.

If an initiative strengthens none of these, it's a feature, not a strategy.



Counter-Positioning

Superior model incumbents can't copy

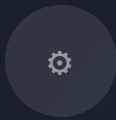
PRIMARY



Switching Costs

Cost of leaving deepens with use

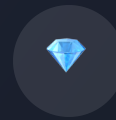
PRIMARY



Process Power

Hard-to-replicate processes

SECONDARY



Cornered Resource

Exclusive access to assets

SECONDARY



Scale Economies

Unit cost declines at volume

DESIGNING



Network Effects

Value grows with users

DESIGNING



Branding

Category ownership = trust

PLANTING

Four reasons incumbents can't copy us.

A single counter-position can be overcome. Four simultaneous ones require the incumbent to execute four architectural rewrites while maintaining their existing product. That is structurally infeasible.



Web-Native SaaS

\$100-\$450/mo vs. \$5K-\$15K/yr desktop licenses

Cameo/DOORS would cannibalize their entire revenue base



AI-Ambient Architecture

Local-first ML pipeline with semantic context contract

Requires rebuilding from the data model up, not a plugin



SysML v2 Native

First-class parser, importer, and diagramizer

Would strand their SysML v1 installed base of 20+ years



Real-Time Collaboration

Yjs/CRDT multiplayer with command→patch pipeline

Would require rewriting their single-user file-locking model

POWER #2: SWITCHING COSTS

The graph deepens. Leaving gets harder.



Traceability Graph

Every trace link, safety goal, coverage metric, and verification evidence accumulates in Luvian's graph. Recreating this in another tool is months of manual work.

Deep by month 6 of usage



Organizational Intelligence

The ML pipeline learns team-specific patterns: modeling style, requirement phrasing, and common failure modes. This intelligence is lost on switching.

Medium by month 12 of usage



Migration Investment

Professional services (\$10–\$50K per engagement) physically move customer data into Luvian. Custom integrations and training are non-transferable.

Immediate at onboarding

Hard to build. Harder to copy.

PROCESS POWER (TAKES YEARS TO REPLICATE)

6-Stage ML Pipeline

Candidate generation → feature extraction → ranking → policy → response shaping → feedback. Domain-tuned, not a GPT wrapper.

3-Layer Rendering Architecture

Semantic → Presentation → React Flow. Enables one-element-many-diagrams. Full rewrite needed to retrofit.

Command → Patch → CRDT

Dispatch pipeline with forward + inverse patches flowing into Y.Doc. Systems-level integration, not a feature.

Semantic Context Contract

All AI consumers share a single versioned context. Prevents the fragmentation that kills AI features in other products.

CORNERED RESOURCES (SECURE NOW)



Patent Portfolio

5 innovations identified. **Must file provisionals before any public demo.**



Domain Training Data

As customers model in Luvian, we accumulate real-world engineering decision graphs, the rarest dataset in MBSE.



Standards Body Access

OMG SysML v2 working group + INCOSE Fellow on advisory board. Influence takes years to build.



Advisory Relationships

Defense procurement, enterprise SaaS, systems engineering. Finite pool, and the first mover gets them.

Gap detection, not suggestion generation.

THE REFRAME

INHERITED STORY

"Ambient ML suggests improvements while you edit"

Competes with GPT wrappers on suggestion quality

FIRST-PRINCIPLES STORY

"Luvian answers: what's missing, what changed, and what's at risk"

Requires the decision graph. Structurally non-replicable by prompt-based competitors

DEFENSIBILITY STACK



Gap Detection

"3 requirements have no verification plan"

Requires Graph



Impact Analysis

"Changing this interface affects 12 test cases"

Requires Graph



Consistency Checking

"These 2 requirements may conflict"

High



Quality Scoring

INCOSE A-F grading at authoring time

Medium



Scaffold Generation

Starter templates from domain knowledge

Low

Invest proportionally to defensibility. Gap detection > suggestions

STRATEGIC POSITIONING

The market has room for multiple winners.

Buyer Archetype	Optimizes For	Best Fit	Luvian's Play
Move-fast hardware	Iteration speed	Flow	Don't compete directly
Requirements-heavy	Traceability, audit	Trace.Space	Compete on scope
Standards-governed	ASIL, INCOSE, certification	LUVIAN	Natural home. Invest here
Defense acquisition	Git-native, MOSA	SysGit	Compete on UX + AI
Early-stage learners	Free, simple	Dalus	Free tier required
Ansys ecosystem	Simulation bundling	Ansys SAM	Target non-Ansys shops

Our segment: Teams building safety-critical systems in regulated industries (automotive OEM, avionics tier 1, defense programs, nuclear/energy) who need integrated architecture + requirements + safety + V&V in one auditable graph.

VS. STARTUP COHORT

Only Luvian covers all four domains.

Capability	Luvian	Flow \$36M	Trace \$11.5M	Dalus YC	SysGit \$6.3M
Architecture modeling (SysML v2)	✓	✗	✗	~	✓
Requirements management	✓	✓	✓	~	✗
Safety analysis (HARA/FMEA/STPA)	✓	✗	✗	✗	✗
V&V / verification evidence	✓	✗	~	✗	✗
INCOSE quality engine	✓	✗	✗	✗	✗
Local-first AI (air-gap)	✓	✗	✗	✗	✓
SOC 2 certification	Not yet	ITAR	SOC2+ISO	Type 2	No

Safety analysis + V&V rows: Luvian is the only startup with coverage. This is the structural moat against the funded cohort.

Enter as the evidence layer. Expand into everything.

Each phase delivers standalone value. Each phase deepens switching costs. No big-bang replacement required.

PHASE 1 • Y1

Evidence Layer

Connect existing tools. See what's missing. Produce compliance evidence automatically.

- ✓ Import from Cameo, DOORS
- ✓ Traceability graph
- ✓ Gap analysis reports

PHASE 2 • Y1-2

Requirements

Author requirements directly with INCOSE quality checking at the point of authoring.

- ✓ INCOSE A-F grading
- ✓ Replaces DOORS
- ✓ Quality engine wedge

PHASE 3 • Y2-3

Modeling

SysML v2 visual modeling with collaboration and AI assistance.

- ✓ Replaces Cameo
- ✓ Real-time collab
- ✓ Ambient ML context

PHASE 4 • Y3+

Safety + V&V

Full decision graph with safety analysis and verification evidence.

- ✓ Replaces spreadsheets
- ✓ Evidence automation
- ✓ Complete graph

ACTIVATION TIMELINE

When each Power activates.

Q2 2026

NOW

Cornered Resource: File patents, recruit advisors, join OMG

Q3 2026

Counter-Positioning: Pitch deck with 4-layer narrative • Whitepaper • SOC 2 initiated

Q4 2026

Switching Costs: 5 design partners • Traceability graphs accumulating • First migrations

Q1 2027

Process + Scale: Seed closed • ML pipeline iterating with real data • First AE

Q2 2027

Branding: INCOSE talk • Gartner briefing • SOC 2 Type 1 • Category narrative in market

2028+

All Powers Compounding: Series A • FedRAMP • Channel partners • 50+ customers

DECISION FRAMEWORK

Five filters for every strategic decision.

1

Does this strengthen a Power?

If not, it's a feature, not a strategy. Features can wait. Powers cannot.

2

Does this serve our natural segment?

Standards-governed teams building safety-critical systems. If it primarily serves Flow's or Dalus's segment, it's a distraction.

3

Does this deepen the graph?

The decision graph is the strategic asset. Every feature should make the graph deeper, richer, or more connected.

4

Is the timing right?

Active now (Counter-Positioning, Switching Costs) = maximum investment. **Designing** (Scale, Network) = architectural investment only.

5

What do first principles say?

Return to the seven axioms: decisions compound, evidence not process, graph answers questions, mutations must be reversible.

THE COMPOUNDING LOOP

**The graph makes the AI better.
The AI makes the graph
more valuable.**



Engineers model

Decisions accumulate in the graph



AI learns patterns

ML pipeline gets smarter with feedback



Gaps detected

Missing traces, conflicts, risks surfaced



Graph deepens

Switching costs compound every cycle

This loop only works with the decision graph.

GPT-wrapper competitors can't build it. They don't have the graph.

SUMMARY

The strategy on one page.

WHO WE ARE

The decision graph engine for safety-critical engineering.

WHO WE SERVE

Teams in regulated industries: automotive, avionics, defense, energy.

HOW WE WIN

4-layer counter-positioning vs. incumbents. Switching costs that deepen with every trace link. Process power in ML and architecture.

HOW WE ENTER

Evidence layer first → requirements → modeling → safety + V&V. Each step delivers value and deepens lock-in.

HOW AI IS OUR MOAT

Not suggestions (anyone wraps GPT). Gap detection: what's missing, what changed, what's at risk. Requires the graph.

WHY WE'RE DIFFERENT

Only product covering all four domains (architecture + requirements + safety + V&V) with local-first AI and air-gap deployment.

\$7–10M seed → 5 design partners → SOC 2 → patents filed → category defined

18 months to prove the graph deepens switching costs as predicted.



Build the graph. Build the moat.

Human-led design. AI-guided modeling.

The platform for safety-critical systems engineering.

Primary	Counter-Positioning + Switching Costs
Secondary	Process Power + Cornered Resource
Designing	Scale Economies + Network Effects + Brand
Core Asset	The Engineering Decision Graph

Luvian Confidential. Internal Strategy